motivating game:

- I observe a variable n times and hide the results to you, then compute mean and variance and let you know only these 2 values
- then a new value of the same variable is observed.

Based on the information you have

1. what do you expect this new value to be ?

2. should you bet on it, how much money would you bet ?

UNIVERSITÀ DEGLI STUDI DELL'AQUILA

Random Variables

what is a random variable?

UNCERTAINTY

It is a variable that can assume different values, for which it is known a probability distribution over such values (states) $\rightarrow$ DOMAIN (ALL POSSIBLE STATES) Ω $\rightarrow$ CONTINUOUS, DISCRETE

probability distribution is how the random variable is distributed over its states when we observe it many time

probability distribution of a random variable measures the "likelihood" of its states

example of a random variable:

$$\text{TOSS A COIN...} P(H) = P(T) = 0.5 = p$$

Ω : {HEAD, TAIL}

BINOMIAL DISTRIBUTION

- Bernoulli process: a sequence of observations of a binary random variable where each observation has probability of success p and of failure $1 - p$ with $p \in (0, 1)$
- Binomial distribution:
  discrete probability distribution of the number of successes k of n independent observations of a Bernoulli process

$$\binom{n}{k} p^k (1 - p)^{n-k}$$

the probability of getting k heads (or tails) by tossing a coin n times

- the probability of n independent events is the product of the probabilities of the single events

$$\prod_{i=1}^{k} p \prod_{i=1}^{n-k} (1-p) = p^k (1-p)^{n-k}$$

- how many possible way of getting k successes on n observations (trials)?

$$\binom{n}{k} = \frac{n!}{k!(n-k)!}$$

example: $k = 2, n = 4 \rightarrow \binom{4}{2} = 6$:

	1	H	H	T	T
	2	T	H	H	T
	3	T	T	H	H
	4	H	T	T	H
	5	H	T	H	T
	6	T	H	T	H

$P_1 = p \cdot p \cdot (1-p) \cdot (1-p)$

$P_2 = (1-p) \cdot p \cdot p \cdot (1-p)$

$\vdots$

P_6

exercise 1.3

$P(2/4) = P_1 \cdot P_2 \cdot P_3 \cdot P_4 \cdot P_5 \cdot P_6$

given a set Ω of all possible values x_i of a discrete random variable X, and a probability p_i associated to each x_i according to a probability distribution

- MEAN,EXPECTED VALUE,PREDICTION

$$\underbrace{m = \mathbb{P}(X) = \sum_{i\in\Omega} p_i x_i}$$

- WEIGHTED SUM OF ALL STATES
- WEIGHTS = PROBABILITIES

DISTRIBUTION OF VALUES

$$\mu = \frac{1}{N} \sum_i x_i =$$

$$= \sum_i \frac{x_i}{N}$$

- VARIANCE

$$var(X) = \sum_{i\in\Omega} p_i (x_i - m)^2$$

DIST. OF VALUES

$$\sigma^2 = \frac{1}{N} \sum_i (x_i - \mu)^2$$

$$= \sum_i \frac{1}{N} (x_i - \mu)^2$$

continuous probability distributions

RANDOM VARIABLES WITH CONTINUOUS DOMAIN

$\Omega = (-\infty, +\infty)$

$\Omega = (0, +\infty)$

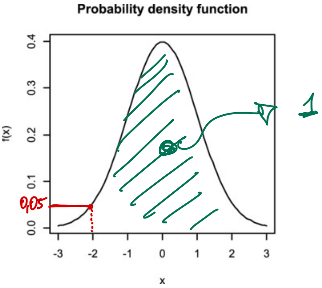
...

- density function $f(x)$:

1) $f(x) \geq 0, f(x) \leq 1,$

2) $\int_{-\infty}^{\infty} f(x)dx = 1$

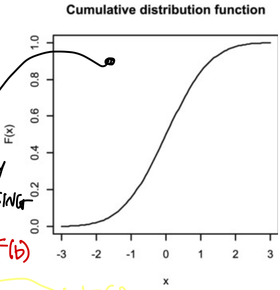
f TAKES AS INPUT A DOMAIN VALUE x
RETURNS AS OUTPUT THE DEGREE OF CONFIDENCE OF x : $f(x)$



- cumulative function $F(x)$:

$$F(x) = \int_{-\infty}^{x} f(y)dy$$

F(x) PROBABILITY OF OBSERVING AT MOST x



MONOTONICALLY NOT DECREASING

what is the probability that...

... $x = a$? $f(a)$

... $x \leq b$? $F(b)$

... $a \leq x \leq b$? $F(b) - F(a)$